Gautam Raghuvanshi

Third Year Undergraduate
Department of Electrical Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - 2025 (Expected)	B.Tech.	Indian Institute of Technology, Kanpur	7.03/10.0
2021	U.P. Board (XII)	Onkareshwar Saraswati Vidya Niketan, Kanpur	92.2%
2019	U.P. Board (X)	Onkareshwar Saraswati Vidya Niketan, Kanpur	97.17%

SCHOLASTIC ACHIEVEMENTS

- Secured **First Position** among more than **30 Lakh** Students in U.P. Board High School Examination 2019 with 97.17% marks.
- Received the **Academic Excellence** Award from honorable Chief Minister of U.P. **Yogi Aditya Nath Ji** and honorable **Ram Nath Kovind ji (Ex-President of India)** for outstanding academic achievements in High school and Intermediate Exams.
- Awarded **Prestegious Ambedkar Ratna Award** from **Governnor of U.P. Ram Naik Ji** for exceptional academic achievements.
- Selected in **Pandit Deen Dayal Memorial Scholarship** given to top meritorious students of U.P. for complete graduation period

RESEARCH EXPERIENCE (VLSI - EDA Laboratory, IIT Kanpur)

Jitter simulation using open-source EDA tools (*May '23 - Ongoing*)

Mentor: ***Prof. Chithra***, Department of Electrical Engineering, IIT Kanpur

Objective	• The project's aim is to figure out how **Jitter simulations** can be done using the open-source EDA tools.
Approach	• Familiarized with the open-source EDA tool **NgSpice**, schematic design interface **Xschem, Skywater130nm pdk**. • Learned fundamental concepts of CMOS inverters working, **delay lines, DLL, PLL, Noises** and **Jitter** . • **Simulated inverter and delay line** using open source Skywater 130nm pdk doing **DC and transient analysis**
Result	• Got the behavior of various simulations performed with different combinations of inputs in an inverter delay Line. • Analyzed the delays occurs in an **Inverters, delay lines** when **different noises** are introduced in the system

KEY PROJECTS

Algorithmic Graph Theory (*Association for Computing Activities, IITK*) (*May '23 - June '23*)

- Explored graph theory from an algorithmic viewpoint. Concepts of **Simple graph traversal algorithms, minimal spanning trees, and the Dijkstra Algorithm** were covered. More complex topics like graph coloring and network flow issues are also covered. .
- Understood and solved interesting problems on **Graph Theory and Ramsey Theory** by Cambridge professor Prof. Timothy Gowers.
- Employed **Kruskal's and Prim's algorithms** for MSTs, **Kosaraju's Tarjan's algorithms** for strongly connected components

Mathematical Trading Strategies (*Stamatics, IITK*) (*May '23 - July '23*)

- Conducted analysis of 10-year historical data for global indices and equities by **applying moving-average strategies**
- Get familiar with **correlation analysis, sophisticated indicators, chart patterns** to generate trade signals and forecast market.
- Technical indicators, such as moving averages, and chart patterns are successfully combined to create an optimal trading system that has been fine-tuned, producing impressive cumulative returns, less drawdown, and a higher Sharpe ratio.

Credit Card Fraud Detection (*Self Project*) (*Dec '22*)

- By working with a high class unbalanced data collection comprising transaction data, fraudulent credit card transactions were discovered.
- Used Machine Learning algorithms including **Logistic Regression, KNN, Random Forest, and Decision Tree** to classify data.
- Plotted the ROC and learning curves of various models, then compared them based on their **F1 score** and **Confusion matrix**.
- Utilizing the **Support Vector Machines (SVM)** technique, a **accuracy of 96 percent** was attained on the testing dataset.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Latex
Familiar: Verilog-HDL, HTML, CSS, Javascript
Utilities: Numpy, Pandas, Sci-Kit Learn, Matplotlib
Software: AUTOCAD, VS Code, MATLAB

RELEVANT COURSEWORK

Fundamentals of Computing	Probability and Statistics	Introduction to Electronics
Control System Analysis	Signal System and Networks	Microelectronics
Introduction to Electrical Engineering	Linear Algebra and Differential Equations	Marketing Management

POSITIONS OF RESPONSIBILITY

Assistant Placement Coordinator | Students' Placement Office, IIT Kanpur (*Mar '23 - Present*)

Leadership	• Coordinating internship process of **900+ students** and placements process of **1400+ students** in team of SPO • Spearheading a team of 70+ members for coordinating and facilitating internship and placement drive • Managing a **unified database of 4500+** companies and **Placement and Internship Portal** of all students.
Initiatives	• Introduced a series of **Internship and Placement insights** together with mock tests for the benefit of students. • Successfully conducted **First Career Fest** of IIT Kanpur **Samvardhan** which enhances the engagement of various new startups and firms, generating many opportunities for the students for internships and placements.
Impact	• Enhanced awareness of the preparation material present on the Students Placement Office's official website • **Increased the number of startups and firms by 10+** coming for **internship and placement drive 2023**

EXTRA-CURRICULARS

Social	• Volunteered at **Ranjit Singh Rozi Shiksha Kendra, IIT Kanpur** as Social Media Admin and **tutor** which works for the education and upliftment of **underprivileged children and women** in rural areas of Kanpur.
Cultural	• **Secretary, Hindi Sahitya Sabha**, Conducted various cultural event such as **Hindi Pakhwada, Kalrav, Akshar** • Mentored 20+ students in **General Championship** and represented **HALL 12** in **Tark** competition in **GC** .
Mentorship	• **Student Guide ,Counselling service**, mentored **Six** first-year students academically and emotionally. • Organised **8** days long orientation program for **First Year Freshers** Students with Counselling Service team.